

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

THEORY EXAMINATION

Question Paper

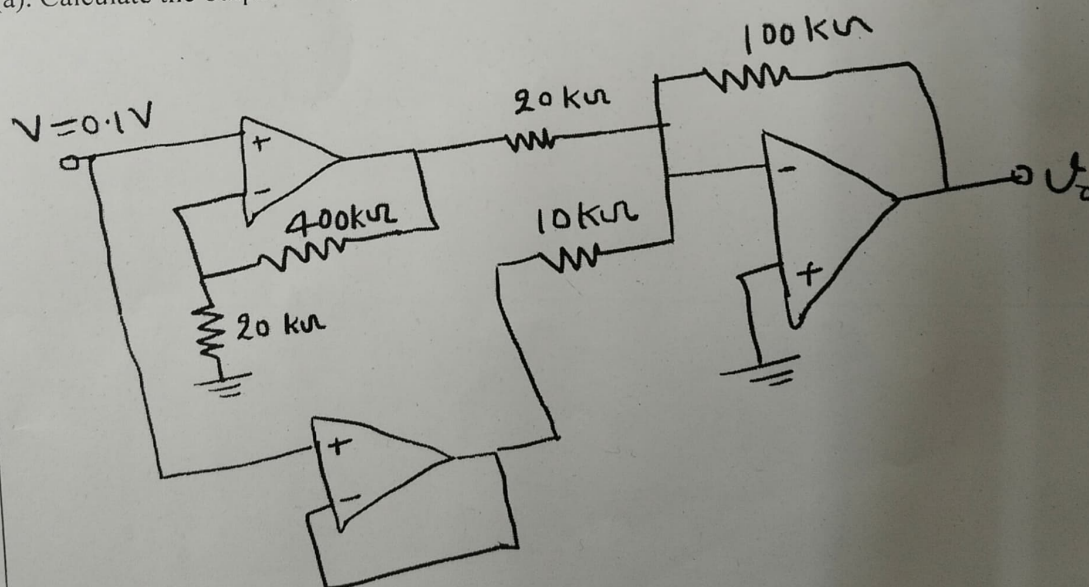
Roll No. 124104069Month and Year of the Examination: Dec.-2025Programme: **B.Tech/M.Tech/MBA/MCA/Ph.D.** Course WorkSemester 3rdSubject: **Analog and Digital Electronics**Course No EEPC-202Maximum Marks 50Number of Questions to be Attempted 05Time allowed 3 hrsTotal No. of Questions 06Total No. of Pages used 02

Unless stated otherwise, the Symbols have their usual meanings in context with the Subject.

Assume suitably and state, additional data required, if any.

The Candidates, before starting to write the solutions, should please check the Question Paper for any discrepancy, and also ensure that they have been delivered the question paper of **right course no.** and right subject title.

1. (a) Explain the concept of Load line and Q-point in BJT. [5]
 (b) In the fixed-bias configuration, $R_C = 2.2 \text{ K}\Omega$, $R_B = 240 \text{ K}\Omega$, $V_{CC} = 12 \text{ V}$, $\beta = 50$, $C_1 = C_2 = 10 \text{ }\mu\text{F}$. With DC Biasing explanation and find out these following I_{BQ} , I_{CQ} , V_{CEQ} , V_B , V_C and V_{BC} . [5]
2. (a). Draw the r_e -equivalent model circuit for a typical voltage-divider bias configuration and derive expression for A_v , Z_i and Z_o . [5]
 (b) In the CE Emitter-Bias Unbypassed (C_E) Configuration network, $R_E = 0.56 \text{ K}\Omega$, $R_C = 2.2 \text{ K}\Omega$, $R_B = 470 \text{ K}\Omega$, $V_{CC} = 20 \text{ V}$, $r_o = 40 \text{ }\Omega$, $\beta = 120$, $C_1 = C_2 = C_E = 10 \text{ }\mu\text{F}$. Find out r_e , Z_i , Z_o and A_v (**Without effect of r_o**). [5]
3. (a). Calculate the output voltage V_o as shown in the figure. [5]



(b) Explain the principle of operation of the RC oscillator. [5]

4 (a) Convert the following:

I. $(378.93)_{10}$ to octal

II. $(5C7)_{16}$ to decimal

III. $(B9F.AE)_{16}$ to octal

IV. $(756.603)_8$ to hexadecimal

V. $(1001.01)_2$ to decimal

[5]

(b) Minimize the following expression using K-map.

I. $f(P, Q, R, S) = \sum m(0, 1, 4, 5, 7, 8, 9, 12, 13, 15)$.

II. $f(P, Q, R, S) = \sum m(0, 3, 7, 10, 14) + \sum d(2, 4, 6, 9, 11, 13)$.

[5]

5. (a) Describe function of full Adder circuit with its truth table, K-map simplification and logic diagram.

[5]

(b) Describe the working of JK Flip-Flop with Truth Table, Characteristic Table, Characteristic Equation, State diagram and Excitation Table.

[5]

6. Write short note any two:

[5*2]

(i) 555 timer

(ii) Multiplexer

(iii) Decoder